

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: <https://www.nature.com/articles/s41586-025-08729-1>.

Note they measure dopamine using fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers changes in fluorescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

1. Using the `pwr` package for R (Champely, 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

Solution: Using `pwr.t.test()`, the researchers would need 21 observations to detect a moderate-to-large effect (as the output was $n = 20.58039$).

```
pwr.t.test(d = 0.65, # moderate-large effect
           power = 0.80,
           sig.level = 0.05,
           alternative = "two.sided",
           type = "one.sample")
```

2. Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the `closer_vals` and `further_vals`. Ensure to `mutate()` the data to get a difference (e.g., `closer_vals - further_vals`).

Solution: To collect the data, we downloaded all of their Figure 2 data, went to the g tabs, then saved just the `closer_vals` and `further_vals` as a separate .csv file to easily use in our code. This was done by simply copying and pasting the two columns together. Mutating the data we also got the difference by subtracting between the two.

```
finch.data = read.csv("./41586_2025_8729_MOESM4_ESM.csv") |>
  mutate(difference = Closer_vals - Farther_vals)
```

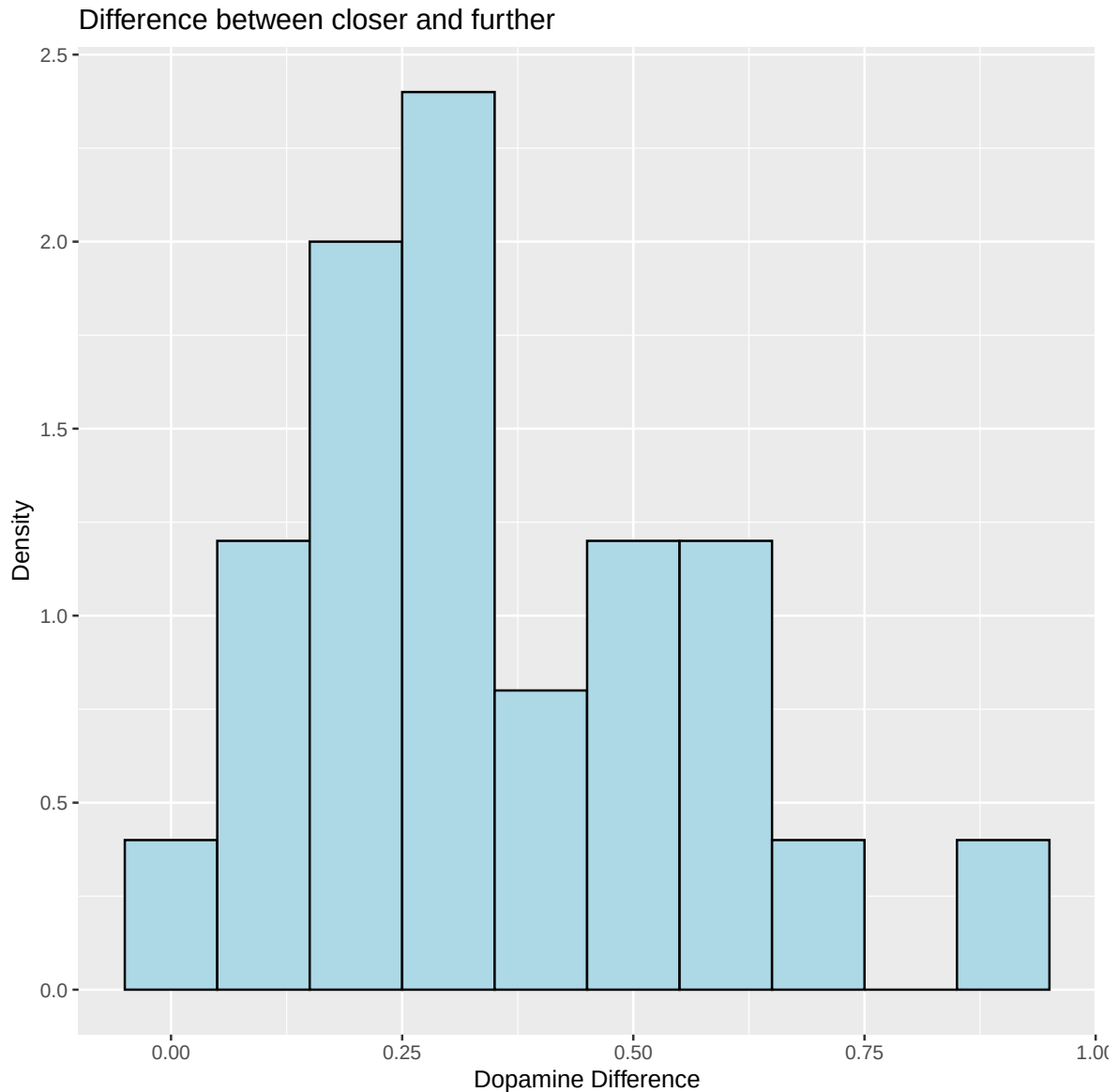
3. Summarize the data.

- (a) Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?
- (b) Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?
- (c) Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?
- (d) **Optional Challenge:** Can you reproduce Figure 2(g)? Note that the you can use `geom_errorbar()` to plot the range created by adding the mean $\pm$ one standard deviation.

Solution: This indicates that the dopamine in the brains of young zebra finches decreases when they sing further from their adult song. Conversely, it increases when they are closer to the adult song, with a suggested difference between the two seen below.

Table 1: Summary of Closer, Further, and the difference in Dopamine

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Closer vals	0.001082	0.089018	0.145534	0.156223	0.195670	0.339488
Further vals	-0.60279	-0.30926	-0.18675	-0.20272	-0.12271	-0.02993
Difference	0.04335	0.22864	0.33208	0.35895	0.46771	0.93188



4. Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60).

Note: Your numbers may vary slightly as they performed some unclear correction of their p -values. I'm waiting to hear back from them via email! **Solution:**

- (a) “The close responses had a large difference from 0 ($p = 8.13 \times 10^{-9}$).”
- (b) “The far responses had a large difference from 0 ($p = 2.59 \times 10^{-8}$).”
- (c) “The difference between populations was large ($p = 1.04 \times 10^{-8}$).”

```
#Closer_vals
closer_t = t.test(x=finch.data$Closer_vals, alternative = "greater", mu = 0)
closer.t.stat = (closer_t[1]$statistic)
interpret_hedges_g(hedges_g(x = finch.data$Closer_vals, alternative = "greater", mu = 0))

#Farther_vals
```

```

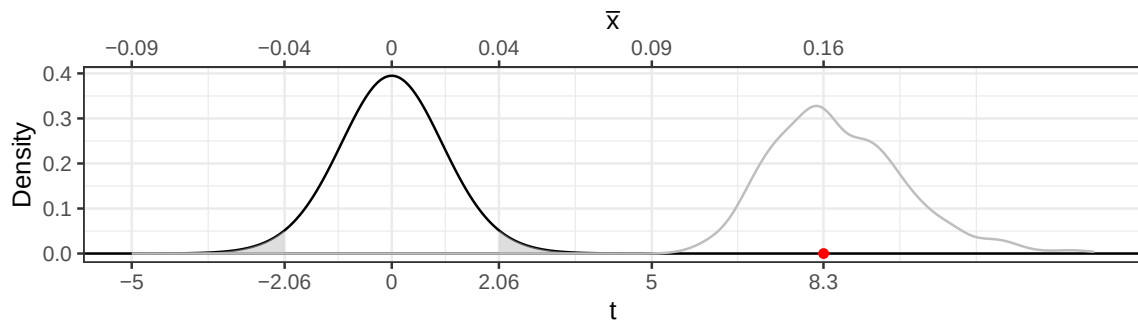
farther_t = t.test(x=finch.data$Farther_vals, alternative = "less", mu = 0)
farther.t.stat = (farther_t[1]$statistic)
interpret_hedges_g(hedges_g(x = finch.data$Farther_vals, alternative = "less", mu = 0))

#Difference
difference_t = t.test(x=finch.data$difference, alternative = "two.sided", mu = 0) #two-sided
diff.t.stat = difference_t[1]$statistic
interpret_hedges_g(hedges_g(x = finch.data$difference, mu = 0)) #two-sided

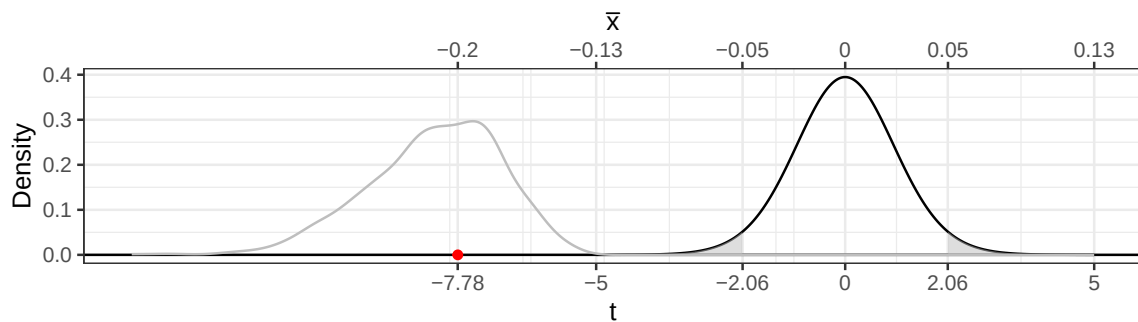
```

- Reverse engineer the hypothesis test plot from Lecture 20 to create accurate hypothesis testing plots for each part of the previous question.

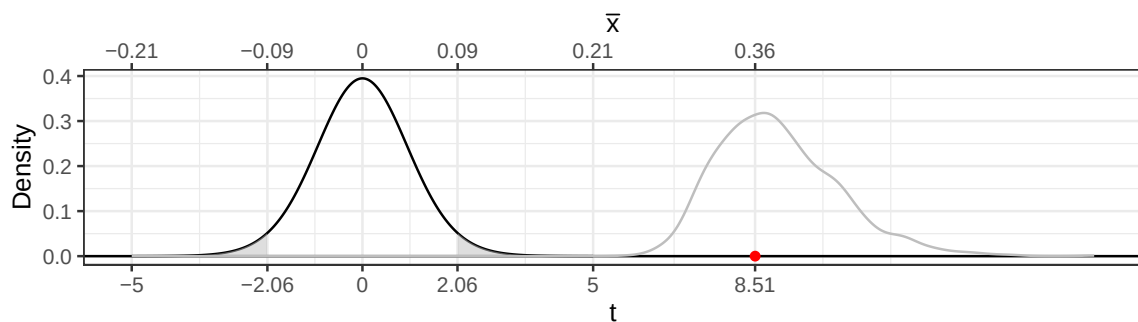
T-Test for Mean Dopamine Closer to Adult Song



T-Test for Mean Dopamine Further from Adult Song



T-Test for Mean Difference in Dopamine



References

Champely, S. (2020). *pwr: Basic Functions for Power Analysis*. R package version 1.3-0.

Kasdin, J., Duffy, A., Nadler, N., Raha, A., Fairhall, A. L., Stachenfeld, K. L., and Gadagkar, V. (2025). Natural behaviour is learned through dopamine-mediated reinforcement. *Nature*, pages 1–8.